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MediScan AI: An Intelligent Medical Record Digitization and Drug-Drug Interaction Warning System Compliance with HL7 FHIR Standards

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Abstract. In the context of Vietnam, a developing country, the risk of handwritten medical prescriptions, with its potential to cause dosage misreading and unmonitored drug-drug interactions (DDIs) among medical fields, still exists. In this paper, we propose MediScan AI, an intelligent microservice-based healthcare system for digitizing handwritten prescriptions and systematically analyzing pharmacology risk. They use Transformer-based OCR with RAG to "debunk" AI hallucinations on medical dosages. The nonstructured extracted data are standardized into international HL7 FHIR format, and coded and mapped by using ATC (Anatomical Therapeutic Chemical) classification system. By accessing a specialized Neo4j graph database that includes both comprehensive drug interaction registries and additional proprietary data or mechanism-based inferred information, MediScan AI provides immediate, tri-level DDI risk alerts. The system complies with local laws (Decree 117/2020/ND-CP) and ensures the data privacy by sophisticated personal identifiable information (PII) anonymization.

Keywords: Medical Record Digitization · Optical Character Recognition · Drug-Drug Interactions · Retrieval-Augmented Generation · HL7 FHIR · Neo4j Graph Database.

1 Introduction

Misprescribing and unaddressed drug-drug interactions (DDIs) represent a massive risk to patient safety worldwide, with the potential to induce adverse drug events (ADEs), extended hospital stays, and increased mortality rates. In the current medical environment in Vietnam, reliance on handwritten medical prescriptions is still exceedingly common, in particular in community clinics and district hospitals. The intrinsic and inheritable nature of the readability issues in clinical handwriting has caused critical drug names, frequencies and dosages misunderstood by patients and pharmacists. In addition, since people routinely see more than one independent specialist, there's no centralized, automated way to check new prescriptions against the records of what they're currently taking. From a legal perspective, the Vietnamese Government promulgated the Decree 117/2020/ND-CP stipulating strict administrative sanctions for illegible or non-conforming writing of prescription forms to regulate systematic digital changes in

clinical settings. Nevertheless, the transformation of unstructured clinical notes to structured and qualified medical fields is facing tremendous challenges in computation. Typical Large Language Models (LLMs) lend towards stochastic hallucinations, which are completely intolerable in high stakes clinical environments where precise chemical ratios are required by law, and biology. For breaking through such hard constraints, we propose **MediScan AI**, a smart, ultra-secure microservices architecture designed for the digitization of handwritten prescription sheets and multi-tiered DDI risk analysis automation. MediScan AI fills a critical challenge between raw unstructured clinical inputs and systemic data validation. The contributions of this work are threefold: (1) An intelligent NLP pipeline based on a customized OCR architecture for medical texts extraction; (2) An end-to-end architectural journey of strict integration of Retrieval-Augmented Generation (RAG) with the international HL7 FHIR namespace to provide full precision alignment; and (3) A graph-driven automated screening engine based on Anatomical Therapeutic Chemical (ATC) codes to compute instant DDI matrix indices.

2 Related Work

Medical record digitization and automated prescription triage have attracted extensive scientific study within the biomedical informatics community. Traditional Optical Character Recognition (OCR) solutions, such as early iterations of Tesseract, exhibited high error rates when processing unsegmented cursive handwriting or low-contrast mobile phone captures. Recent iterations integrating Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have drastically improved Named Entity Recognition (NER) in clinical contexts, yet mapping these raw extracted text blocks into standardized, interoperable health records remains an active research challenge.

Concurrently, automated drug interaction monitoring has shifted from static relational databases towards semantic knowledge structures. Relational database management systems (RDBMS) encounter significant performance bottlenecks when processing complex, multi-layered chemical pathways or reciprocal dependencies across hundreds of active ingredients. Research indicates that graph databases, such as Neo4j, offer superior traversal efficiency for interconnected network models, enabling recursive path-finding algorithms to detect latent drug contradictions.

Nevertheless, existing commercial tools often fail to bridge the semantic divide between raw language extractions and clinical terminology constraints. Most packages operate as isolated systems, requiring medical staff to manually select structured checkboxes rather than extracting variables seamlessly from a casual smartphone photograph. Furthermore, strict formatting compatibility with global frameworks like the Health Level Seven Fast Healthcare Interoperability Resources (HL7 FHIR) is rarely incorporated natively into consumer-facing mobile solutions. MediScan AI differentiates itself by unifying non-hallucinatory

RAG frameworks, strict international standardization protocols, and high-performance graph processing engines into an integrated, compliant infrastructure.

3 Methodology

The technical architecture of MediScan AI is engineered upon a secured, modular microservices topology, precisely separating data ingestion, deep learning inference, and structured graph processing.

3.1 Intelligent Ingestion and Vision-Based Document OCR

The workflow initiates when the patient transmits an image of a handwritten prescription sheet via the client interface. The ingestion service performs pre-processing operations, including adaptive binarization, skew correction, and contrast normalization. The core extraction engine utilizes a specialized Transformer-based vision model fine-tuned on cursive alphanumeric handwriting patterns. The resulting extracted sequence undergoes Named Entity Recognition (NER) utilizing a clinical BERT variant to isolate three vital medical primitives: active chemical agents, precise dosage metrics, and administration frequencies.

3.2 Non-Hallucinatory RAG Architecture and HL7 FHIR Standardization

To prevent catastrophic factual errors regarding medication dosages, the system completely isolates the core LLM from free-form generation by implementing a strict **Retrieval-Augmented Generation (RAG)** pipeline. Instead of relying on internal parameter weights, the LLM acts purely as a semantic compiler. The extracted OCR primitives are checked against a verified, local drug compendium index.

Once verified, the unstructured text is automatically mapped into an immutable JSON representation adhering to the international **HL7 FHIR** (Fast Healthcare Interoperability Resources) schema. Specifically, data fields are serialized into a standard **MedicationRequest** resource structure, mapping local pharmacy terms directly to global conceptual registries such as RxNorm.

3.3 ATC-Coded Knowledge Graph and Multi-Tiered DDI Alert Engine

The localized drug data points are converted into standard **Anatomical Therapeutic Chemical (ATC)** classification indices. MediScan AI maintains an active knowledge engine built upon a **Neo4j Graph Database**, where nodes symbolize active chemical categories (mapped by ATC codes) and edges denote known clinical drug-drug interactions derived from verified pharmaceutical registries.

When a new prescription is processed, the system constructs a transactional query matrix, intersecting the newly extracted ATC nodes with the patient's

existing "Digital Health Log" repository. The graph engine evaluates interaction paths and categorizes risks into three distinct notification tiers:

1. **Critical Alert (Severe Risk):** Immediate contraindications that endanger life; triggers persistent UI blocks.
2. **Cautionary Alert (Moderate Risk):** Requires specific clinical monitoring or potential adjustments in dosage timing.
3. **Safe Status (No Interaction):** Complements the user's historical active database without conflicting chemical indicators.

3.4 Legal Compliance and Security Protocols

To align with the data confidentiality expectations of **Decree 117/2020/ND-CP**, a dedicated security gateway processes all records before cloud persistency. A Personal Identifiable Information (PII) anonymization layer separates all patient identity elements (names, identity numbers) from clinical diagnosis strings, storing them in separate, encrypted relational nodes. All data objects are subjected to AES-256 encryption at rest, with SSL/TLS transport layer cryptography protecting all external API transactions.

4 References

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